

Killian **QUETEL**

Mechanical & Surface Design Engineer · CATIA R&D

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PROFILE

Mechanical engineer with **6+ years** across aerospace, space and CAD software R&D (Dassault Systèmes, Airbus, DCUBED, Safran). I manage the portfolio of **CATIA core engineering roles** and work on **Body-in-White (BIW)** with many automotive customers, turning industrial needs into R&D roadmaps, and I keep my hands on the product: high-quality surfacing, robust parametric and **modular design**, generative design and simulation-driven engineering. My flagship work is a complete **2026 F1 car designed from scratch** in CATIA.

EXPERIENCE

CATIA R&D Core Engineering Roles Portfolio Manager
Apr 2024 – present

Dassault Systèmes · Vélizy-Villacoublay, France

- Own the portfolio and roadmap of CATIA core design roles (part, surface, assembly) on 3DEXPERIENCE, from customer needs to prioritized R&D requirements.
- Work on **Body-in-White (BIW)** design with many customers in the automotive industry, from their engineering processes to CATIA requirements.
- Specify, validate and launch new capabilities with R&D teams, including AI-assisted design (**Command Intelligence**), working in an Agile (LeSS) organization.
- Build industrial demonstrators on real products: **F1 virtual twin** (front wings 2019 & 2025, brake air duct entry), CubeSat, wheel rims, generative door and hood.
- Present to customers and the community: CATIA Talk speaker, F1 video series featured by the official CATIA page.

Mechanical System Engineer (VIE via Alten)
Jul 2023 – Jan 2024

DCUBED · Munich, Germany

- Owned the deployment mechanisms of a multi flat-panel reflectarray antenna for small satellites (**ESA ARTES – MARAS project**).
- Covered every step: R&D, simulation & validation, manufacturing and test campaign; co-author of the related **IAC 2024** paper.

Mechanical Engineer & CAD Process Engineer (VIE via Alten)
2022 – 2023

Airbus · Hamburg & Munich, Germany

- **A350 cabin:** modeling of galley sets in CATIA V5 (4 months).
- **MDO CAD process:** topology optimization support on 3DEXPERIENCE and export control compliance (6 months).

Apprentice Engineer – CATIA R&D
Sep 2020 – Sep 2022

Dassault Systèmes · Vélizy-Villacoublay, France

- **Generative design:** demonstrators coupling structural and fluid topology optimization, subdivision modeling, city- and bridge-scale models by reverse engineering (Bir-Hakeim bridge: from generative design to steel structure).
- **Sustainable campus:** energy-harvesting mechanisms (speed bump, turnstile, piezo floor), simulated in SIMULIA and Dymola, with a 1:20 working 3D-printed model.

Production Quality Assurance – Apprentice
2019 – 2020

Safran Aircraft Engines · France

- Quality control on turbine-blade lines (M88 Rafale, LEAP, CFM56), reducing costs and lead times.

SKILLS

CAD & surfacing

CATIA 3DEXPERIENCE · GSD+ · Part Design · Assembly Design · Body-in-White · Modular design · Imagine & Shape · Generative Design · CATIA V5 · SolidWorks · Fusion 360

Simulation

SIMULIA (CFD, structural) · Abaqus · Ansys · HyperMesh · OptiStruct · Dymola

Product & management

Portfolio & roadmap · Requirements · Agile (LeSS) · Customer demos

Code & tools

Python · C# · MATLAB · Arduino · 3D printing · Adobe CC

EDUCATION

MSc Engineering – ESILV
Computational mechanics & digital modeling · 2018 – 2022 · work-study

BSc Science – Paris-Saclay
1st year, materials & aerospace · 2017 – 2018

HIGHLIGHTS

IAC 2024, Milan – co-author
Design & deployment feasibility study of a 3-panel reflectarray antenna for a 12U CubeSat

CATIA Talk – speaker
Live LinkedIn presentation in English, 150+ viewers

3DEXPERIENCE certifications (2022)
Function Driven Generative Designer · Product Manager – Associate

LANGUAGES

French	Native
English	C1
Spanish	B1
German	A2

BEYOND WORK

- French Army reservist since 2019
- MMA, BJJ, boxing · marathons & trail · cycling · handball
- Motorsport, photography, mountains & travel